

Matías G. Delgadino

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Education

2011–2016 Ph.D. in Mathematics, University of Maryland, College Park, USA.
Thesis: Analysis of Self-Organization. *Advisor:* Antoine Mellet.

2006–2011 Licenciado en Matemática, Universidad Nacional de Córdoba, Argentina.
Thesis: Teoría de Control aplicada a tratamientos de quimioterapia
Advisor: Andres Barrea.

Academic Positions

2021–Present Assistant Professor, Department of Mathematics, The University of Texas at Austin.

2020–2021 Hooke Research Fellow (on leave from PUC-Rio), University of Oxford, United Kingdom.

2019–2021 Assistant Professor, PUC-Rio, Brazil.

2017–2019 Postdoctoral Researcher, Imperial College London, United Kingdom.

2016–2017 Postdoctoral Researcher, ICTP, Trieste, Italy.

Publications

*Numbered in reverse chronological order. Preprints are listed first. **M.G. Delgadino** denotes the candidate; * indicates mentored student or postdoc.*

Preprints and Work Under Review

[27] F.U. Caja-Lopez*, **M.G. Delgadino**, J. Kitagawa. Stability of optimal transport maps and second variation of the 2-Monge-Kantorovich distance. Preprint, 2026. [arXiv:2605.24232](https://arxiv.org/abs/2605.24232). *Under review at ASENS.*

[26] **M.G. Delgadino**, S.A. Smith. Mass generation for the two-dimensional $O(N)$ linear sigma model in the large N limit. Preprint, 2026. [arXiv:2601.19630](https://arxiv.org/abs/2601.19630). *Under review at Communications on Pure and Applied Mathematics.*

- [25] **M.G. Delgadino**, S. Motsch, A. Parulekar, W. Porteous*, S. Shakkottai. Diffusion-based posterior sampling: a Feynman–Kac analysis of bias and stability. Preprint, 2026. [arXiv:2605.06538](#). *Under review at NeurIPS*.
- [24] **M.G. Delgadino**, L. Melzi*. Quantitative stability of the Rossby–Haurwitz waves of degree two for the Euler equation on $\mathbb{S}^2$. Preprint, 2025. [arXiv:2509.16156](#). *Under review at SIAM Journal on Mathematical Analysis*.
- [23] **M.G. Delgadino**, R.S. Gvalani. Sharp mean-field estimates for the repulsive log gas in any dimension. Preprint, 2025. [arXiv:2506.22083](#). *Under review at Annals of Probability*.
- [22] F.U. Caja-Lopez*, **M.G. Delgadino**, M.P. Gualdani, M. Taskovic. Contractivity of Wasserstein distance and exponential decay for the Landau equation with Maxwellian molecules. Preprint, 2025. [arXiv:2504.13802](#). *Under review at Analysis & PDE*.

Published and Accepted

- [21] M. Coti Zelati, **M.G. Delgadino**. Entropy maximization in the two-dimensional Euler equations. *Analysis & PDE*, 2026. [arXiv:2405.14738](#).
- [20] **M.G. Delgadino**, M. Vaughan*. Continuous symmetrizations and uniqueness of solutions to nonlocal equations. *Analysis & PDE*, 2025. [arXiv:2310.13221](#).
- [19] **M.G. Delgadino**, B.B. Suassuna*, R. Cabrera*. Generative adversarial networks: dynamics. *Journal of Machine Learning Research* **26**(181), 1–30, 2025. [arXiv:2405.18673](#).
- [18] J.A. Carrillo, **M.G. Delgadino**, L. Desvillettes, J.S.H. Wu*. The Landau equation as a gradient flow. *Analysis & PDE* **17**(4), 1331–1375, 2024. [arXiv:2007.08591](#).
- [17] **M.G. Delgadino**, D. Weser*. A Heintze–Karcher inequality with free boundaries and applications to capillarity theory. *Journal of Functional Analysis* **287**(9), 110584, 2024. [arXiv:2210.16376](#).
- [16] **M.G. Delgadino**, R.S. Gvalani, G.A. Pavliotis, S.A. Smith. Phase transitions, logarithmic Sobolev inequalities, and uniform-in-time propagation of chaos for weakly interacting diffusions. *Communications in Mathematical Physics* **401**(1), 275–323, 2023. [arXiv:2112.06304](#).
- [15] J.A. Carrillo, **M.G. Delgadino**, J.S.H. Wu*. Convergence of a particle method for a regularized spatially homogeneous Landau equation. *Mathematical Models and Methods in Applied Sciences* **33**(5), 971–1008, 2023. [arXiv:2211.07015](#).
- [14] J.A. Carrillo, **M.G. Delgadino**, J.S.H. Wu*. Boltzmann to Landau from the gradient flow perspective. *Nonlinear Analysis* **219**, 112824, 2022. [arXiv:2107.07252](#).

- [13] J.A. Carrillo, **M.G. Delgadino**, R.L. Frank, M. Lewin. Fast diffusion leads to partial mass concentration in Keller–Segel type stationary solutions. *Mathematical Models and Methods in Applied Sciences* **32**(4), 831–850, 2022. [arXiv:2012.08586](#).
- [12] **M.G. Delgadino**, X. Yan, Y. Yao. Uniqueness and nonuniqueness of steady states of aggregation-diffusion equations. *Communications on Pure and Applied Mathematics* **75**(1), 3–59, 2022. [arXiv:1908.09782](#).
- [11] **M.G. Delgadino**, R.S. Gvalani, G.A. Pavliotis. On the diffusive-mean field limit for weakly interacting diffusions exhibiting phase transitions. *Archive for Rational Mechanics and Analysis* **241**(1), 91–148, 2021. [arXiv:2001.03920](#).
- [10] **M.G. Delgadino**, A. Mellet. On the relationship between the thin film equation and Tanner’s law. *Communications on Pure and Applied Mathematics* **74**(3), 507–543, 2021. [arXiv:1901.09611](#).
- [9] J.A. Carrillo, **M.G. Delgadino**, G.A. Pavliotis. A λ -convexity based proof for the propagation of chaos for weakly interacting stochastic particles. *Journal of Functional Analysis* **279**(10), 108734, 2020. [arXiv:1906.04601](#).
- [8] M. Coti Zelati, **M.G. Delgadino**, T.M. Elgindi. On the relation between enhanced dissipation time-scales and mixing rates. *Communications on Pure and Applied Mathematics* **73**(6), 1205–1244, 2020. [arXiv:1806.03258](#).
- [7] J.A. Carrillo, **M.G. Delgadino**, J. Dolbeault, R.L. Frank, F. Hoffmann. Reverse Hardy–Littlewood–Sobolev inequalities. *Journal de Mathématiques Pures et Appliquées* **132**, 133–165, 2019. [arXiv:1807.09189](#).
- [6] J.A. Carrillo, **M.G. Delgadino**, F.S. Patacchini. Existence of ground states for aggregation-diffusion equations. *Analysis and Applications* **17**(3), 393–423, 2019. [arXiv:1803.01915](#).
- [5] **M.G. Delgadino**, F. Maggi. Alexandrov’s theorem revisited. *Analysis & PDE* **12**(6), 1613–1642, 2019. [arXiv:1711.07690](#).
- [4] **M.G. Delgadino**. Convergence of a one-dimensional Cahn–Hilliard equation with degenerate mobility. *SIAM Journal on Mathematical Analysis* **50**(4), 4457–4482, 2018. [arXiv:1510.05021](#).
- [3] **M.G. Delgadino**, S.A. Smith. Hölder estimates for fractional parabolic equations with critical divergence free drifts. *Annales de l’Institut Henri Poincaré C* **35**(3), 577–604, 2018. [arXiv:1609.02691](#).
- [2] **M.G. Delgadino**, F. Maggi, C. Mihaila, R. Neumayer. Bubbling with L^2 -almost constant mean curvature and an Alexandrov-type theorem for crystals. *Archive for Rational Mechanics and Analysis* **230**(3), 1131–1177, 2018. [arXiv:1705.10117](#).

- [1] J.A. Carrillo, **M.G. Delgadino**, A. Mellet. Regularity of local minimizers of the interaction energy via obstacle problems. *Communications in Mathematical Physics* **343**(3), 747–781, 2016. [arXiv:1406.4040](#).

Interdisciplinary Work

- [I.3] U.A. Trigos-Raczkowski, R. Lyons, **M.G. Delgadino**, A.S. Ackleh, A. Ostling. Disturbance-generated competitive coexistence. Preprint, 2023. [bioRxiv:2023.11.02.565332](#). *Theoretical ecology / population dynamics*.
- [I.2] F.N. Cámara Halac, **M.G. Delgadino**. DreamSound: deep activation layer sonification. *Proceedings of the International Computer Music Conference (ICMC 2021)*, Georgia Institute of Technology, 2021. [\[full text\]](#). *Computer music / deep-learning sonification*.
- [I.1] L.D. Simonelli, **M.G. Delgadino**, F.N. Cámara Halac. Hearing the Self: A Spectral Experience. *Proceedings of the International Computer Music Conference (ICMC 2017)*, Shanghai, China, 2017. [\[website\]](#). *Computer music / spectral sound performance*.

Grants and Awards

Submitted

- 2026–2031 NSF CAREER: *Singular Particle Systems and Diffusion Models* (**pending**). \$500,000.
- 2026–2029 NSF-DMS Standard Grant: *Interacting Diffusions with Singular Potentials: Theory and Applications to Generative Modeling* (**pending**). \$280,000.

Awarded

- 2022–2026 NSF-DMS Standard Grant: *Optimal Transport Applications to Probability, Machine Learning, and Kinetic Theory*. \$220,000.
- 2020 Rita Levi Montalcini Award (declined). €195,000. Italy.
- 2020–2021 Young Scientist Support, Instituto Serrapilheira. R\$ 100,000. Brazil.
- 2020–2023 Bolsas de Produtividade em Pesquisa, CNPq. R\$ 40,000. Brazil.

Ph.D. Students

- 2026–Present **Jakob Wellington**, UT Austin. Co-advised with Prof. F. Maggi.
- 2025–Present **Luca Melzi**, Imperial College London. Co-advised with Prof. M. Coti Zelati.
- 2025–Present **Francisco Unai Caja-Lopez**, UT Austin. Co-advised with Prof. M.P. Gualdani.
- 2023–Present **Will Porteous**, UT Austin. Co-advised with Prof. I.M. Gamba.
- 2021–2025 **Kenneth DeMason**, UT Austin. Currently postdoc at McMaster University.
- 2020–2022 **Daniel Weser**, UT Austin. Co-advised with Prof. F. Maggi. Currently postdoc at UNC.
- 2018–2022 **Jeremy Wu**, University of Oxford. Co-advised with Prof. J.A. Carrillo. Postdoc at UCLA; currently Assistant Professor at the University of Manitoba.

Postdoctoral Mentoring

- 2022–2025 **Rene Cabrera**, Postdoc supported by the Analysis RTG, UT Austin. Joint publication in *Journal of Machine Learning Research*. Currently Instructor at UT Austin.
- 2021–2023 **Mary Vaughan**, Postdoc supported by the Analysis RTG, UT Austin. Joint publication in *Analysis & PDE*. Postdoc at University of Western Australia; currently Assistant Professor at Texas State University.

Undergraduate Mentoring

- 2022–2024 **Reese Feldmeier**, Dean’s Honors Program, UT Austin. REU funded by the Analysis RTG; undergraduate research and senior thesis advisor. Currently Ph.D. student at Stanford.
- 2019–2021 **Bruno Suassuna** PUC-Rio, Brazil. Currently Ph.D. student at PUC-Rio.
- 2019–2021 **Humberto Seghetto** [PICME](#), PUC-Rio, Brazil. Currently Ph.D. student at PUC-Rio.

Conference and Workshop Organization

- Fall 2025 Organizing Committee, [SIAM Conference on Analysis of Partial Differential Equations \(PD25\)](#).
- Spring 2024 Main Organizer, [Winter School in Machine Learning](#), UT Austin.
- 2019 Main Organizer, UNESCO/ICTP 1st Latin American School in Applied Mathematics. UFSQ/EPN, Quito, Ecuador. [\[video\]](#).

Professional Service

Grant Panels NSF-DMS Standard Panel (2023); NSF Mathematical Foundations of AI Panel (2025, 2026).

Journals Inventiones Mathematicae; Journal of the European Mathematical Society; Archive for Rational Mechanics and Analysis; Communications in Mathematical Physics; Analysis & PDE; ESAIM: Control, Optimisation and Calculus of Variations; Journal of Functional Analysis; SIAM Journal on Mathematical Analysis; Probability Theory and Related Fields; Annals of Probability; Annales de l'Institut Henri Poincaré C; Transactions of the American Mathematical Society; Nonlinearity; Mathematical Models and Methods in Applied Sciences.

Invited Talks

Talks are grouped by topic; within each group, listed in reverse chronological order.

— *On Diffusion Posterior Sampling.*

May 2026 Imperial College, London, UK

May 2026 OxPDE, University of Oxford, Oxford, UK

March 2026 University of Arizona, Tucson, USA.

March 2026 North Carolina State University, USA.

— *Sharp estimates for the log-gas.*

November 2025 Minisymposium: Advances in PDE Modeling and Analysis for Life and Social Sciences, SIAM PD25, Pittsburgh, USA.

October 2025 Calderón–Zygmund Seminar, University of Chicago, USA.

August 2025 National University of Singapore, Singapore.

May 2025 Center for Nonlinear Analysis, Carnegie Mellon University, Pittsburgh, USA.

— *Generative Adversarial Networks: Dynamics and Mode Collapse.*

August 2024 Banff–Oaxaca Workshop, Oaxaca, Mexico.

July 2024 AMS–UMI Joint Meeting Mini-symposium, Palermo, Italy.

May 2024 Texas Differential Equations Conference (TXDE), Texas State University, San Marcos, USA.

April 2024 Calderón–Zygmund Seminar, University of Chicago, USA.

April 2024 CIRM Workshop, Marseille, France.

April 2024 VAPS Online Seminar (hosted by UCLA).

March 2024 Seminar, Vanderbilt University, USA.

January 2024 Conference on Numerical Methods for Optimal Transport Problems, Mean Field Games, and Multi-Agent Dynamics, Universidad Técnica Federico Santa María, Valparaíso, Chile.

— *Entropy maximization in the two-dimensional Euler equations.*

December 2023 Workshop on Recent Advances in Fluid Dynamics: Singularity, Regularity and Mixing, Duke–Kunshan University, China.

December 2023 Seminar, Chinese Academy of Sciences, China.

— *Boltzmann to Landau from the gradient flow perspective.*

November 2023 BIRS: Recent Progress in Kinetic and Integro-Differential Equations, Banff, Canada (online).

— *Phase transitions, logarithmic Sobolev inequalities, and uniform-in-time propagation of chaos for weakly interacting diffusions.*

July 2023 Max Planck Institute for Mathematics, Bonn, Germany.

July 2023 Workshop on Interacting Particle Systems, Imperial College London, UK.

March 2023 Applied Mathematics and PDE Seminar, University of Maryland, College Park, USA.

March 2023 PDE Seminar, Simon Fraser University, Canada.

November 2022 BIRS: Kinetic Equations, Oaxaca, Mexico.

March 2022 PDE Seminar, National University of Singapore.

March 2022 Workshop: Frontiers in the Interplay Between Probability and Kinetic Theory, ICMS, Edinburgh, UK.

February 2022 Frontiers in Kinetic Theory (KineCon 2022), Newton Institute, Cambridge, UK.

— *Rearrangement for higher-order models.*

June 2023 BIRS–IMAG: Nonlinear Diffusion and Nonlocal Interaction Models, Granada, Spain.

— *Mixing and enhanced dissipation.*

August 2019 Brazilian Mathematics Colloquium: PDE Session, IMPA, Rio de Janeiro, Brazil.

July 2019 Analysis Seminar, ICTP, Trieste, Italy.

February 2019 Colloquium, ShanghaiTech University, China.

February 2019 Analysis and PDE Seminar, University of Bath, UK.

December 2018 Fluid Mechanics Seminar, Imperial College London, UK.

November 2018 Fluid Mechanics Seminar, University of Warwick, UK.

— *Propagation of chaos by Gamma-convergence.*

June 2019 Workshop on Optimal Transport in Analysis and Probability, Erwin Schrödinger Institute (ESI), Vienna, Austria.

May 2019 Workshop: Probabilistic and Variational Methods in Kinetic Theory, Hausdorff Research Institute for Mathematics, Bonn, Germany.

— *On the relationship between the thin film equation and Tanner’s law.*

January 2019 PDE Seminar, Georgia Tech, USA.

October 2018 Applied Analysis Seminar, MPI Leipzig, Germany.

— *Alexandrov’s theorem revisited.*

October 2018 KCL/UCL Geometry Seminar, King’s College London, UK.

July 2018 Calculus of Variations and Geometric Measure Theory Conference, University of Sussex, UK.

March 2018 Analysis Seminar, University of Texas at Austin, USA.

March 2018 PDE / Applied Mathematics Seminar, University of Maryland, College Park, USA.

February 2018 Geometric Analysis Seminar, University of Sussex, UK.

January 2018 PDE and Geometric Analysis Seminar, University of Cambridge, UK.

— *Interaction energy.*

October 2017 PDE Seminar, Imperial College London, UK.

September 2017 Analysis Seminar, University of Texas at Austin, USA.

July 2017 Workshop on Aggregation-Diffusion PDEs, Anacapri, Italy.

May 2017 Mathematical Analysis and Applications Seminar, Università di Pavia, Italy.

— *The relationship between the obstacle problem and minimizers of the interaction energy.*

October 2016 Workshop: Interactions between Partial Differential Equations and Functional Inequalities, Mittag-Leffler Institute, Sweden.

April 2016 Differential Equations Seminar, University of Tennessee, Knoxville, USA.

March 2016 Nonlinear Analysis and PDEs Seminar, City University of New York, USA.

June 2015 6th Symposium on Analysis and PDEs, Purdue University, USA.